# MySQL Assignment 2 -Querying Data

## Queries demonstrating Clauses & Operators

### 1. DISTINCT VALUES:

* A query to retrieve distinct salaries from the Employees table.

```
SELECT DISTINCT salary
FROM employees;
```

| salary |
|---|
| 60000.00 |
| 80000.00 |
| 45000.00 |
| 70000.00 |
| 55000.00 |
| 65000.00 |
| 90000.00 |
| 95000.00 |
| 75000.00 |
| 85000.00 |
| 100000.00 |

### 2. ALIAS (AS):

* Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

```
SELECT age AS employee_age FROM employees;
```

| employee_age |
|---|
| 27 |
| 31 |
| 26 |
| 29 |
| 25 |
| 28 |
| 32 |
| 27 |
| 30 |
| 29 |
| 28 |
| 26 |
| 33 |
| 30 |
| 27 |
| 31 |
| 28 |
| 25 |
| 29 |
| 32 |
| 26 |

```
SELECT salary AS employee_salary FROM employees;
```

| employee_salary |
|---|
| 60000.00 |
| 80000.00 |
| 45000.00 |
| 70000.00 |
| 55000.00 |
| 65000.00 |
| 90000.00 |
| 70000.00 |
| 55000.00 |
| 80000.00 |
| 60000.00 |
| 95000.00 |
| 75000.00 |
| 60000.00 |
| 85000.00 |
| 100000.00 |
| 80000.00 |
| 75000.00 |
| 55000.00 |
| 90000.00 |
| 85000.00 |

**3. WHERE CLAUSE & OPERATORS:**

* Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

```
SELECT * FROM employees
WHERE salary > 50000 AND hire_date < '2016-01-01' ;
```

| employee_id | employee_name | gender | age | hire_date | designation | department_id | location_id | salary |
|---|---|---|---|---|---|---|---|---|
| 5001 | Vihaan Singh | M | 27 | 2015-01-20 | Data Analyst | 3 | 4 | 60000.00 |
| 5002 | Reyansh Singh | M | 31 | 2015-03-10 | Network Engineer | 12 | 1 | 80000.00 |
| 5004 | Kiara Malhotra | F | 29 | 2015-07-05 | NULL | 8 | 3 | 70000.00 |
| 5005 | Anvi Chaudhary | F | 25 | 2015-09-11 | Business Development Executive | 11 | 1 | 55000.00 |
| 5006 | Dhruv Shetty | M | 28 | 2015-11-20 | UI Developer | 8 | 2 | 65000.00 |

* Find the employee whose designation is missing and fill it with "Data Scientist".

```
SELECT * FROM employees
WHERE designation IS NULL;
```

| employee_id | employee_name | gender | age | hire_date | designation | department_id | location_id | salary |
|---|---|---|---|---|---|---|---|---|
| 5004 | Kiara Malhotra | F | 29 | 2015-07-05 | NULL | 8 | 3 | 70000.00 |

```
UPDATE employees
SET designation = 'Data Scientist'
WHERE designation IS NULL;
```

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 5004 | Kiara Malhotra | F | 29 | 2015-07-05 | Data Scientist | 8 | 3 | 70000.00 |

## Sorting and Grouping Data:

### 1. ORDER BY:

● Find employees sorted by department ID in ascending order and salary in descending order.

```
SELECT * FROM employees
ORDER BY department_id ASC, salary DESC;
```

| employee_id | employee_name | gender | age | hire_date | designation | department_id | location_id | salary |
|---|---|---|---|---|---|---|---|---|
| 5020 | Aarav Nair | M | 32 | 2018-03-05 | Software Engineer | 1 | 1 | 90000.00 |
| 5007 | Anushka Singh | F | 32 | 2016-01-15 | Marketing Manager | 2 | 3 | 90000.00 |
| 5023 | Anika Paul | F | 30 | 2018-09-20 | Public Relations Specialist | 2 | 2 | 70000.00 |
| 5011 | Saanvi Patel | F | 28 | 2016-09-20 | Marketing Analyst | 2 | 1 | 60000.00 |
| 5025 | Avni Iyengar | F | 31 | 2019-01-20 | Data Scientist | 3 | 4 | 100000.00 |
| 5026 | Vivaan Singh | M | 29 | 2019-03-10 | Business Analyst | 3 | 2 | 75000.00 |
| 5001 | Vihaan Singh | M | 27 | 2015-01-20 | Data Analyst | 3 | 4 | 60000.00 |
| 5016 | Zoya Khan | F | 31 | 2017-07-05 | Legal Counsel | 4 | 4 | 100000.00 |
| 5022 | Aadhya Iyengar | F | 28 | 2018-07-10 | HR Specialist | 4 | 4 | 60000.00 |
| 5019 | Ishika Patel | F | 29 | 2018-01-15 | Talent Acquisition Specialist | 4 | 4 | 55000.00 |
| 5024 | Aryan Shetty | M | 27 | 2018-11-05 | Product Manager | 5 | 1 | 95000.00 |
| 5030 | Sara Iyer | F | 28 | 2019-11-20 | Project Manager | 5 | 1 | 90000.00 |
| 5027 | Ananya Paul | F | 32 | 2019-05-20 | Content Writer | 6 | 3 | 60000.00 |
| 5028 | Anaya Kapoor | F | 26 | 2019-07-05 | Event Coordinator | 6 | 1 | 60000.00 |
| 5015 | Ishaan Kumar | M | 27 | 2017-05-20 | Financial Analyst | 7 | 1 | 85000.00 |
| 5021 | Advik Kapoor | M | 26 | 2018-05-20 | Finance Analyst | 7 | 3 | 85000.00 |

### 2. LIMIT:

● Display the first 5 employees hired in the year 2018.

```
SELECT * FROM employees
WHERE hire_date >= '2018-01-01' AND hire_date <='2018-12-31'
LIMIT 5;
```

| employee_id | employee_name | gender | age | hire_date | designation | department_id | location_id | salary |
|---|---|---|---|---|---|---|---|---|
| 5019 | Ishika Patel | F | 29 | 2018-01-15 | Talent Acquisition Specialist | 4 | 4 | 55000.00 |
| 5020 | Aarav Nair | M | 32 | 2018-03-05 | Software Engineer | 1 | 1 | 90000.00 |
| 5021 | Advik Kapoor | M | 26 | 2018-05-20 | Finance Analyst | 7 | 3 | 85000.00 |
| 5022 | Aadhya Iyengar | F | 28 | 2018-07-10 | HR Specialist | 4 | 4 | 60000.00 |
| 5023 | Anika Paul | F | 30 | 2018-09-20 | Public Relations Specialist | 2 | 2 | 70000.00 |

### 3. AGGREGATE FUNCTIONS:

● Calculate the sum of all salaries in the Finance department.

```
SELECT SUM(salary) AS Finance_Total_salary
FROM employees
WHERE department_id = (
    SELECT department_id
    FROM departments
    WHERE department_name='Finance'
);
```

| Finance_Total_salary |
|---|
| 170000.00 |

● Find the minimum age among all employees.

```
SELECT MIN(age) AS min_age
FROM employees ;
```

| min_age |
|---|
| 25 |

## Grouping data using GROUP BY

● List the maximum salary for each location.

```
SELECT
    (SELECT location
    FROM location l
    WHERE l.location_id = e.location_id) AS locations,
    MAX(salary) AS max_salary
FROM employees e
GROUP BY e.location_id ;
```

| locations | max_salary |
|---|---|
| Chennai | 95000.00 |
| Bangalore | 95000.00 |
| Hyderabad | 90000.00 |
| Pune | 100000.00 |

● Calculate the average salary for each designation containing the word 'Analyst'.

```
SELECT designation , AVG(salary) AS avg_salary
FROM employees
WHERE designation LIKE '% Analyst'
GROUP BY designation
```

| designation | avg_salary |
|---|---|
| Data Analyst | 60000.000000 |
| Marketing Analyst | 60000.000000 |
| Supply Chain Analyst | 60000.000000 |
| Financial Analyst | 85000.000000 |
| Finance Analyst | 85000.000000 |
| Business Analyst | 75000.000000 |
| Quality Assurance Analyst | 80000.000000 |

## Filtering data using HAVING

● Find departments with less than 3 employees.

```
SELECT d.department_name, COUNT(e.employee_id) AS employee_count
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
GROUP BY d.department_name
HAVING employee_count < 3 ;
```

| department_name | employee_count |
|---|---|
| Business Development | 2 |
| Content Creation | 2 |
| Finance | 2 |
| Operations | 1 |
| Product Management | 2 |
| Research and Development | 1 |
| Software Development | 1 |

● Find locations with female employees whose average age is below 30.

```
SELECT l.location, ROUND(AVG(e.age)) AS female_avg_age
FROM employees e
JOIN location l
ON l.location_id = e.location_id
WHERE gender = 'F'
GROUP BY l.location
HAVING female_avg_age < 30;
```

| location | female_avg_age |
|---|---|
| Bangalore | 27 |
| Chennai | 27 |
| Pune | 29 |

## Join Queries

**1. INNER JOIN:**

● List employee names, their designations, and department names where employees are assigned to a department.

```
SELECT e.employee_name, e.designation, d. department_name
FROM employees e
INNER JOIN departments d
ON e.department_id = d.department_id
WHERE e.department_id IS NOT NULL ;
```

| employee_name | designation | department_name |
|---|---|---|
| Aaradhya Iyer | Customer Support Executive | Customer Support |
| Arnav Rao | Customer Success Manager | Customer Support |
| Vihaan Mohan | Supply Chain Analyst | Customer Support |
| Vihaan Singh | Data Analyst | Data Science |
| Avni Iyengar | Data Scientist | Data Science |
| Vivaan Singh | Business Analyst | Data Science |
| Kiara Malhotra | Data Scientist | Design |
| Dhruv Shetty | UI Developer | Design |
| Diya Jha | Graphic Designer | Design |
| Ishaan Kumar | Financial Analyst | Finance |
| Advik Kapoor | Finance Analyst | Finance |
| Zoya Khan | Legal Counsel | Human Resources |
| Ishika Patel | Talent Acquisition Specialist | Human Resources |
| Aadhya Iyengar | HR Specialist | Human Resources |
| Reyansh Singh | Network Engineer | IT |
| Atharv Yadav | Systems Administrator | IT |
| Kabir Nair | IT Support Specialist | IT |
| Arjun Kumar | Quality Assurance Analyst | IT |
| Anushka Singh | Marketing Manager | Marketing |
| Saanvi Patel | Marketing Analyst | Marketing |
| Anika Paul | Public Relations Specialist | Marketing |
| Myra Verma | Operations Manager | Operations |
| Aryan Shetty | Product Manager | Product Management |
| Sara Iyer | Project Manager | Product Management |
| Ishan Mishra | Research Scientist | Research and Develo... |

**2. LEFT JOIN:**

● List all departments along with the total number of employees in each department, including departments with no employees.

```
SELECT d.department_name, COUNT(e.employee_id) AS total_employees
FROM departments d
LEFT JOIN employees e
ON e.department_id = d.department_id
GROUP BY d.department_name;
```

| department_name | total_employees |
|---|---|
| Business Development | 2 |
| Content Creation | 2 |
| Customer Support | 3 |
| Data Science | 3 |
| Design | 3 |
| Finance | 2 |
| Human Resources | 3 |
| IT | 4 |
| Marketing | 3 |
| Operations | 1 |
| Product Management | 2 |
| Research and Development | 1 |
| Software Development | 1 |

**3. RIGHT JOIN:**

● Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

```
SELECT l.location, IFNULL(e.employee_name, "NULL") AS emp_name
FROM employees e
RIGHT JOIN location l
ON l.location_id = e.location_id;
```

| location | emp_name |
|---|---|
| Bangalore | Aaradhya Iyer |
| Bangalore | Dhruv Shetty |
| Bangalore | Myra Verma |
| Bangalore | Vihaan Mohan |
| Bangalore | Kabir Nair |
| Bangalore | Anika Paul |
| Bangalore | Vivaan Singh |
| Bangalore | Arjun Kumar |
| Chennai | Reyansh Singh |
| Chennai | Anvi Chaudhary |
| Chennai | Saanvi Patel |
| Chennai | Ishaan Kumar |
| Chennai | Aarav Nair |
| Chennai | Aryan Shetty |
| Chennai | Anaya Kapoor |
| Chennai | Sara Iyer |
| Hyderabad | Kiara Malhotra |
| Hyderabad | Anushka Singh |
| Hyderabad | Kiaan Desai |
| Hyderabad | Arnav Rao |
| Hyderabad | Ishan Mishra |
| Hyderabad | Advik Kapoor |
| Hyderabad | Ananya Paul |
| Pune | Vihaan Singh |

**4. CROSS JOIN:**

● Show all possible combinations of departments and locations.

```
SELECT d.department_name, l.location
FROM departments d
CROSS JOIN location l;
```

| department_name | location |
|---|---|
| Business Development | Pune |
| Business Development | Hyderabad |
| Business Development | Chennai |
| Business Development | Bangalore |
| Content Creation | Pune |
| Content Creation | Hyderabad |
| Content Creation | Chennai |
| Content Creation | Bangalore |
| Customer Support | Pune |
| Customer Support | Hyderabad |
| Customer Support | Chennai |
| Customer Support | Bangalore |
| Data Science | Pune |
| Data Science | Hyderabad |
| Data Science | Chennai |
| Data Science | Bangalore |
| Design | Pune |
| Design | Hyderabad |
| Design | Chennai |
| Design | Bangalore |
| Finance | Pune |
| Finance | Hyderabad |

**5. SELF JOIN:**

- Show pairs of employees working in the same department, excluding self-pairs.

```
SELECT e1.employee_name AS emp1, e2.employee_name AS emp2, e1.department_id
FROM employees e1
JOIN employees e2
ON e1.department_id = e2.department_id
AND e1.employee_id < e2.employee_id;
```

| emp1 | emp2 | department_id |
|---|---|---|
| Vihaan Singh | Avni Iyengar | 3 |
| Vihaan Singh | Vivaan Singh | 3 |
| Reyansh Singh | Atharv Yadav | 12 |
| Reyansh Singh | Kabir Nair | 12 |
| Reyansh Singh | Arjun Kumar | 12 |
| Aaradhya Iyer | Arnav Rao | 10 |
| Aaradhya Iyer | Vihaan Mohan | 10 |
| Kiara Malhotra | Dhruv Shetty | 8 |
| Kiara Malhotra | Diya Jha | 8 |
| Anvi Chaudhary | Kiaan Desai | 11 |
| Dhruv Shetty | Diya Jha | 8 |
| Anushka Singh | Saanvi Patel | 2 |
| Anushka Singh | Anika Paul | 2 |
| Atharv Yadav | Kabir Nair | 12 |
| Atharv Yadav | Arjun Kumar | 12 |
| Saanvi Patel | Anika Paul | 2 |
| Arnav Rao | Vihaan Mohan | 10 |
| Ishaan Kumar | Advik Kapoor | 7 |
| Zoya Khan | Ishika Patel | 4 |
| Zoya Khan | Aadhya Iyen... | 4 |
| Kabir Nair | Arjun Kumar | 12 |
| Ishika Patel | Aadhya Iyen... | 4 |

## Windows function

● Write a window function query to rank employees by salary using **rank().**

```
SELECT
    employee_name,
    salary,
    RANK() OVER(ORDER BY salary) AS rank_status
FROM employees;
```

| employee_name | salary | rank_status |
|---|---|---|
| Aaradhya Iyer | 45000.00 | 1 |
| Anvi Chaudhary | 55000.00 | 2 |
| Kiaan Desai | 55000.00 | 2 |
| Ishika Patel | 55000.00 | 2 |
| Vihaan Singh | 60000.00 | 5 |
| Saanvi Patel | 60000.00 | 5 |
| Vihaan Mohan | 60000.00 | 5 |
| Aadhya Iyengar | 60000.00 | 5 |
| Ananya Paul | 60000.00 | 5 |
| Anaya Kapoor | 60000.00 | 5 |
| Dhruv Shetty | 65000.00 | 11 |
| Kiara Malhotra | 70000.00 | 12 |
| Diya Jha | 70000.00 | 12 |
| Anika Paul | 70000.00 | 12 |
| Arnav Rao | 75000.00 | 15 |
| Ishan Mishra | 75000.00 | 15 |
| Vivaan Singh | 75000.00 | 15 |
| Reyansh Singh | 80000.00 | 18 |
| Atharv Yadav | 80000.00 | 18 |
| Kabir Nair | 80000.00 | 18 |

● Write a window function query to rank employees by salary within each department using **DENSE_RANK()**

```
SELECT
    employee_name,
    salary,
    department_id,
    DENSE_RANK() OVER(PARTITION BY department_id ORDER BY salary DESC) AS rank_status
FROM employees;
```

| employee_name | salary | department_id | rank_status |
|---|---|---|---|
| Aarav Nair | 90000.00 | 1 | 1 |
| Anushka Singh | 90000.00 | 2 | 1 |
| Anika Paul | 70000.00 | 2 | 2 |
| Saanvi Patel | 60000.00 | 2 | 3 |
| Avni Iyengar | 100000.00 | 3 | 1 |
| Vivaan Singh | 75000.00 | 3 | 2 |
| Vihaan Singh | 60000.00 | 3 | 3 |
| Zoya Khan | 100000.00 | 4 | 1 |
| Aadhya Iyengar | 60000.00 | 4 | 2 |
| Ishika Patel | 55000.00 | 4 | 3 |
| Aryan Shetty | 95000.00 | 5 | 1 |
| Sara Iyer | 90000.00 | 5 | 2 |
| Ananya Paul | 60000.00 | 6 | 1 |
| Anaya Kapoor | 60000.00 | 6 | 1 |
| Ishaan Kumar | 85000.00 | 7 | 1 |
| Advik Kapoor | 85000.00 | 7 | 1 |
| Kiara Malhotra | 70000.00 | 8 | 1 |
| Diya Jha | 70000.00 | 8 | 1 |
| Dhruv Shetty | 70000.00 | 8 | 2 |
| Ishan Mishra | 75000.00 | 9 | 1 |
| Arnav Rao | 75000.00 | 10 | 1 |
| Vihaan Mohan | 60000.00 | 10 | 2 |

- Write a window function query, **Running total** salary by department

```
SELECT employee_name, department_id, salary,
    SUM(salary)
    OVER(PARTITION BY department_id ORDER BY salary) AS running_total
FROM employees;
```

| employee_name | department_id | salary | running_total |
|---|---|---|---|
| Aarav Nair | 1 | 90000.00 | 90000.00 |
| Saanvi Patel | 2 | 60000.00 | 60000.00 |
| Anika Paul | 2 | 70000.00 | 130000.00 |
| Anushka Singh | 2 | 90000.00 | 220000.00 |
| Vihaan Singh | 3 | 60000.00 | 60000.00 |
| Vivaan Singh | 3 | 75000.00 | 135000.00 |
| Avni Iyengar | 3 | 100000.00 | 235000.00 |
| Ishika Patel | 4 | 55000.00 | 55000.00 |
| Aadhya Iyengar | 4 | 60000.00 | 115000.00 |
| Zoya Khan | 4 | 100000.00 | 215000.00 |
| Sara Iyer | 5 | 90000.00 | 90000.00 |
| Aryan Shetty | 5 | 95000.00 | 185000.00 |
| Ananya Paul | 6 | 60000.00 | 120000.00 |
| Anaya Kapoor | 6 | 60000.00 | 120000.00 |
| Ishaan Kumar | 7 | 85000.00 | 170000.00 |
| Advik Kapoor | 7 | 85000.00 | 170000.00 |
| Dhruv Shetty | 8 | 65000.00 | 65000.00 |
| Kiara Malhotra | 8 | 70000.00 | 205000.00 |